Indore Institute of Science & Technology

Department of Artificial Intelligence & Machine Learning

MINOR PROJECT - DESIGN DIAGRAMS

NyayMitra: An AI-Powered Multilingual Legal Guidance System using Retrieval-Augmented Generation and Case Classification

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Note: These diagrams describe the complete NyayMitra design. The frontend dashboard (SOS, helplines, accessibility menu, legal notice generator, explore, directory and news pages) is implemented; the AI engine (case classifier, RAG retriever, explanation and draft generation, FastAPI backend) is the planned work and is shaded differently in the diagrams that show both.

Fig. 1: System Architecture Diagram

Layered design of NyayMitra: the frontend prototype (green) calls a FastAPI layer, which drives the AI core (language detection, case classifier, confidence check, RAG retriever, explanation and draft generators) backed by the legal corpus, vector index, labelled dataset and database. Planned components are shaded.

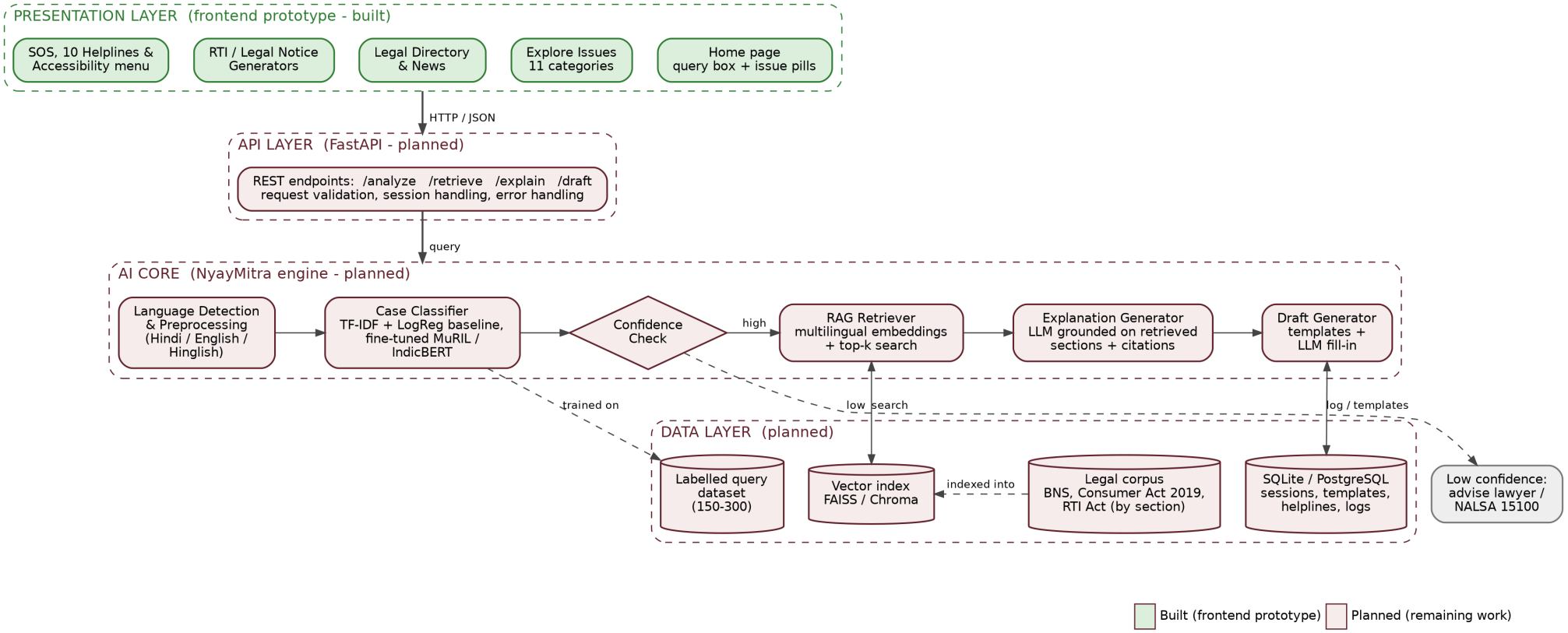


Fig. 2: System Flowchart

End-to-end flow from a user's query to guidance and a draft document, including the emergency (SOS) branch and the low-confidence branch that recommends a lawyer or NALSA free legal aid instead of answering with false confidence.

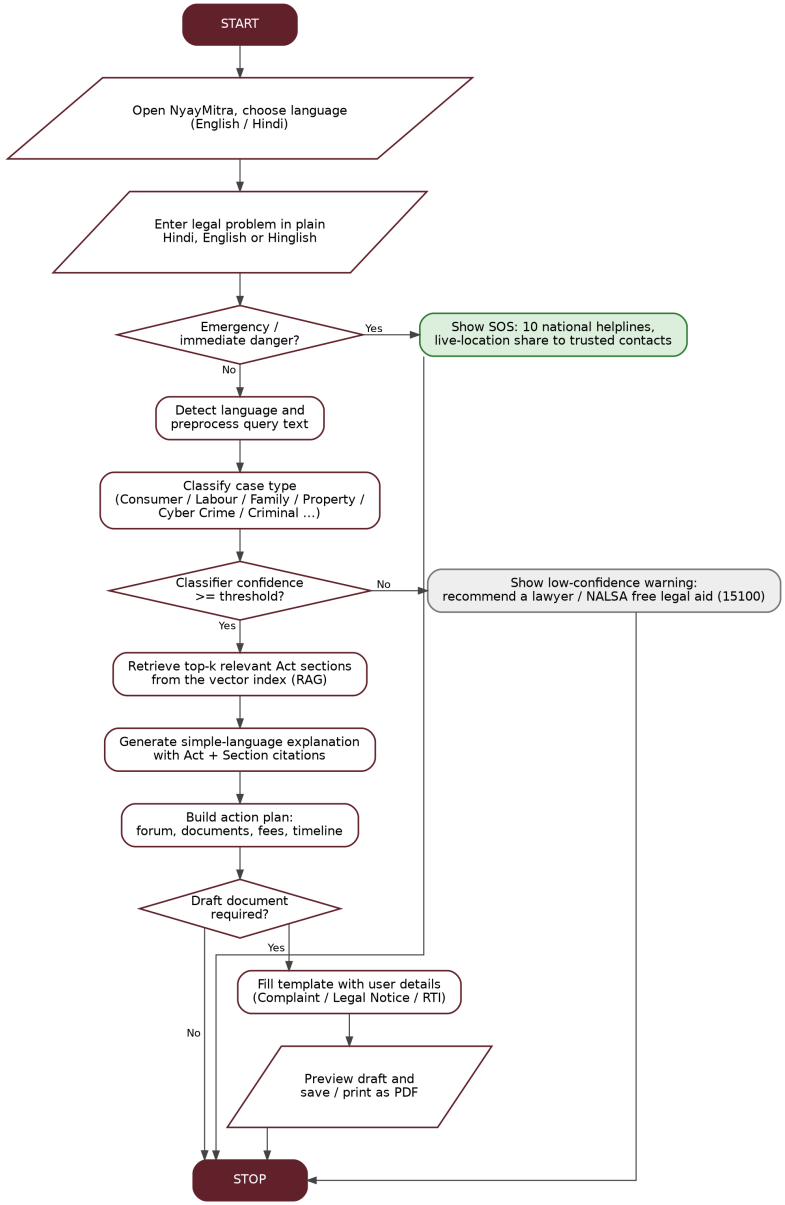


Fig. 3: Data Flow Diagram - Level 0 (Context Diagram)

NyayMitra as a single process with its external entities: the Citizen, the Administrator who maintains legal content, and the official legal sources the corpus is compiled from.

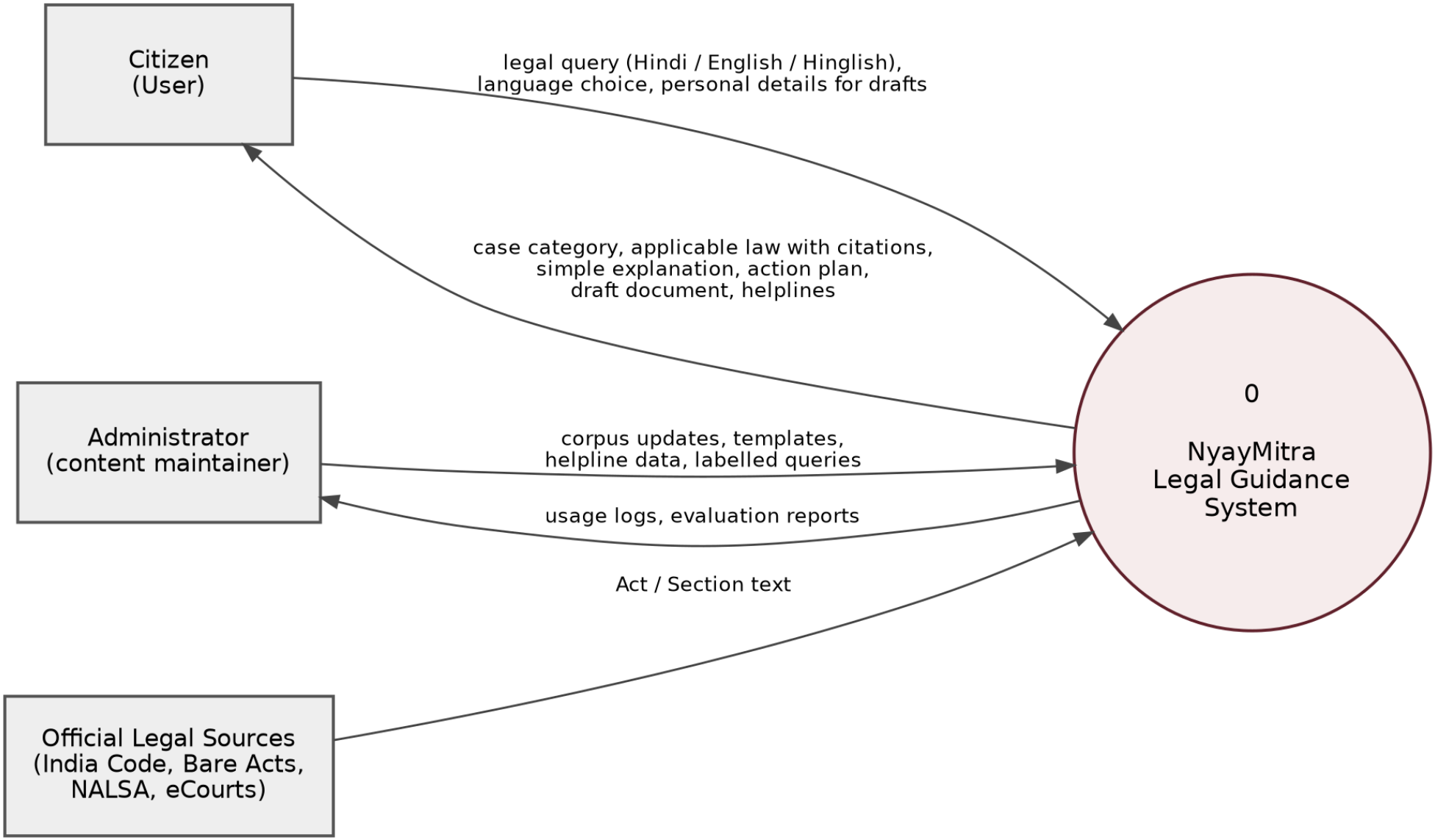


Fig. 4: Data Flow Diagram - Level 1

Decomposition into five processes (preprocess, classify, retrieve, explain with action plan, generate draft) and four data stores (query log, trained classifier and labelled data, legal corpus with vector index, templates and helplines).

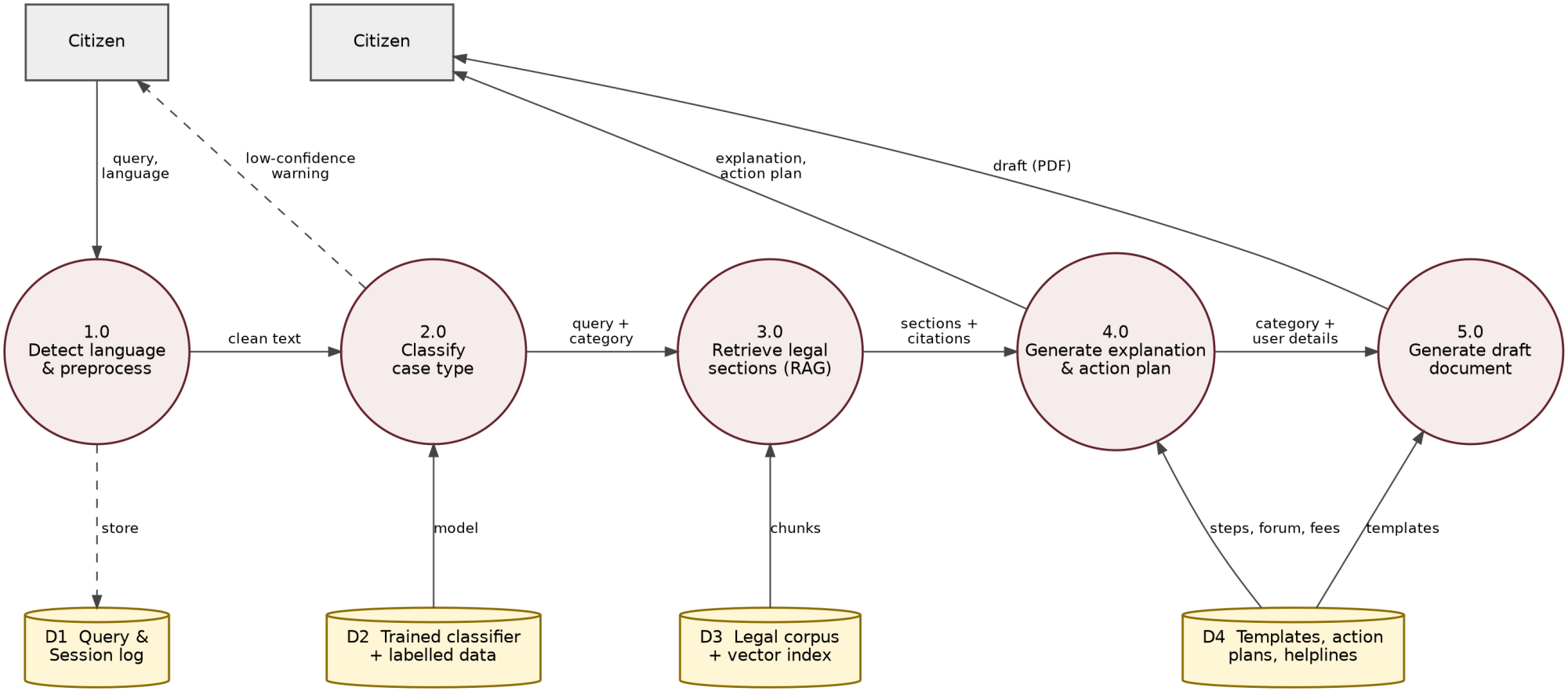


Fig. 5: Use Case Diagram

Citizen use cases (query, action plan, drafts, SOS, directory, settings) and Administrator use cases (corpus, templates, classifier retraining). UC1 includes classification (UC2) and retrieval (UC3); the LLM service is a secondary actor.

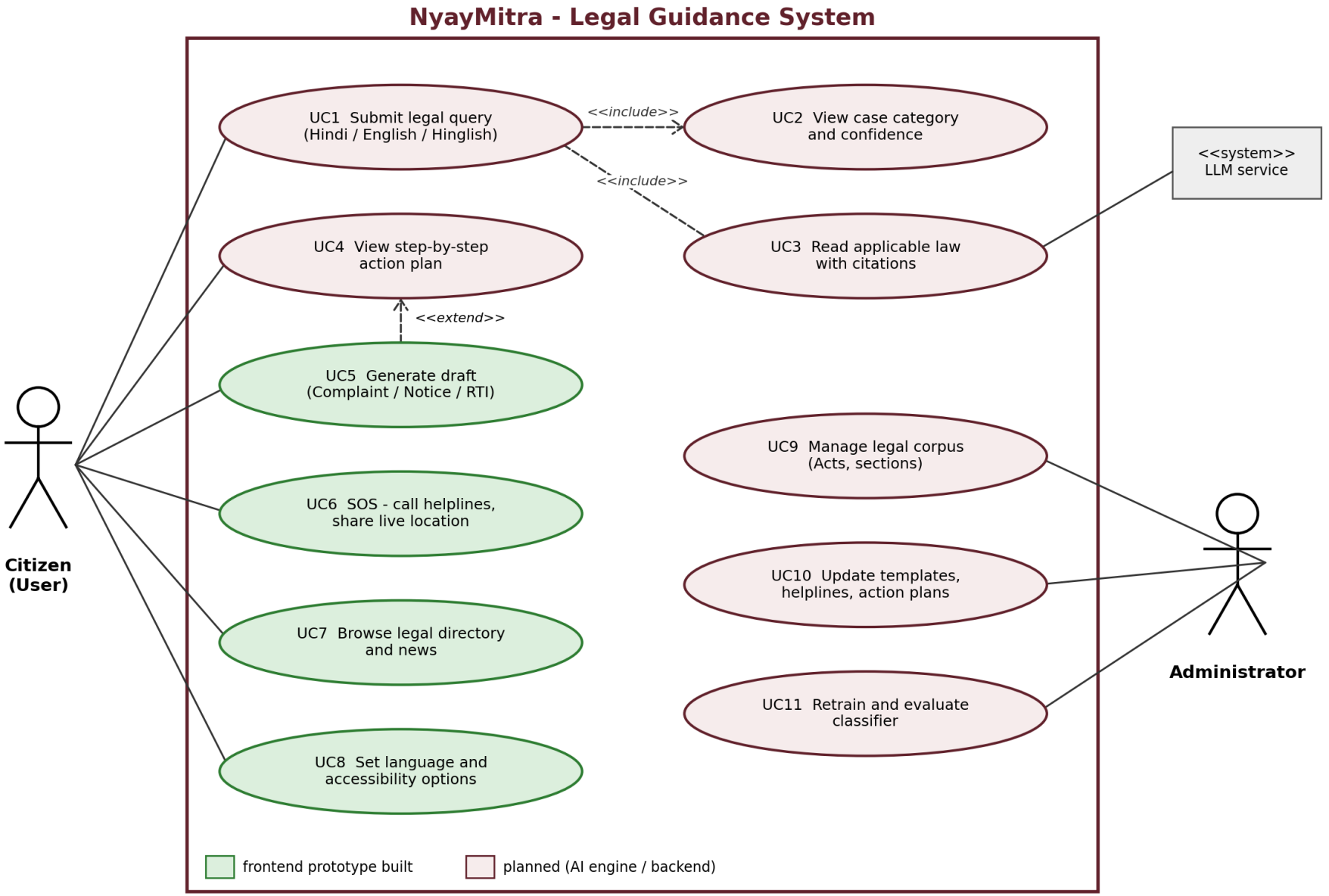


Fig. 6: Entity-Relationship Diagram

Proposed SQLite / PostgreSQL schema: sessions and queries, classification results, case categories with action steps, helplines and draft templates, the legal Act / Section corpus metadata, retrieval results and generated drafts.

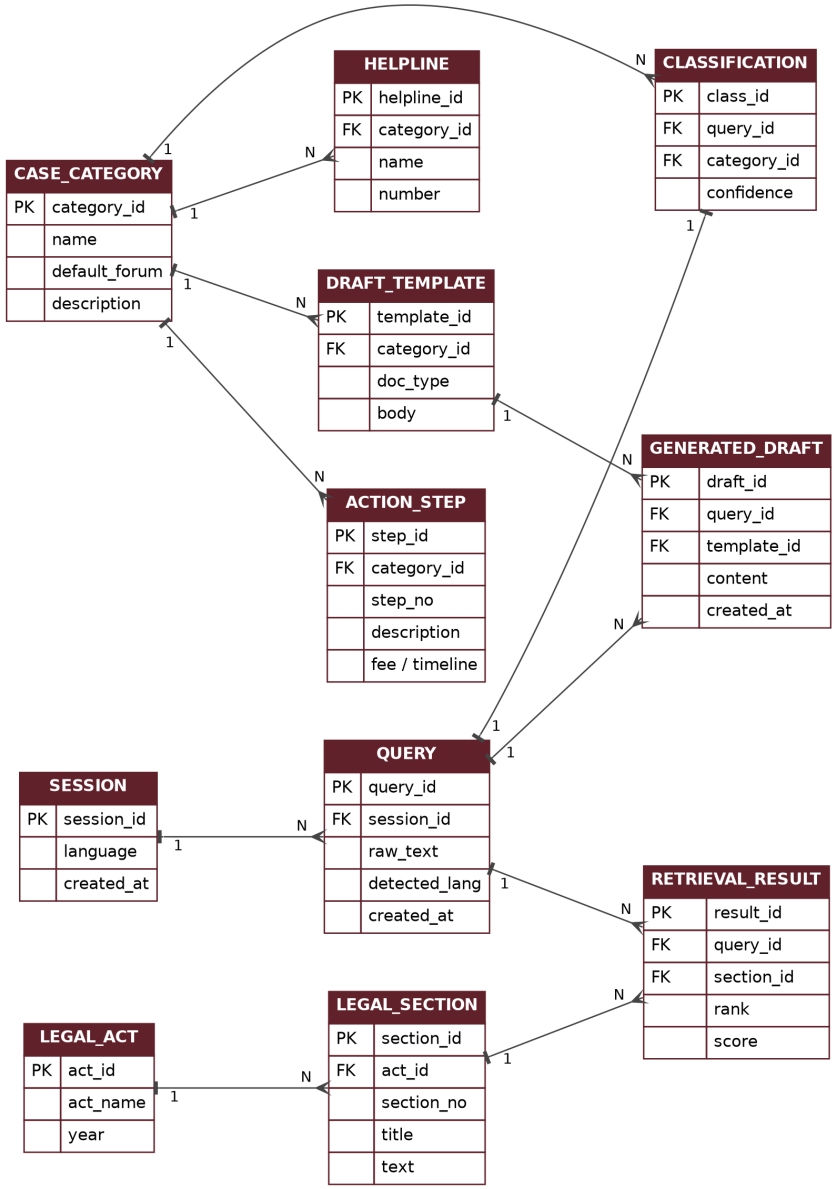


Fig. 7: ML / RAG Pipeline (Methodology Diagram)

Three pipelines: (A) knowledge-base preparation and vector indexing, (B) classifier training and evaluation (TF-IDF + Logistic Regression baseline vs fine-tuned MuRIL / IndicBERT), and (C) live inference that combines the saved model and the index.

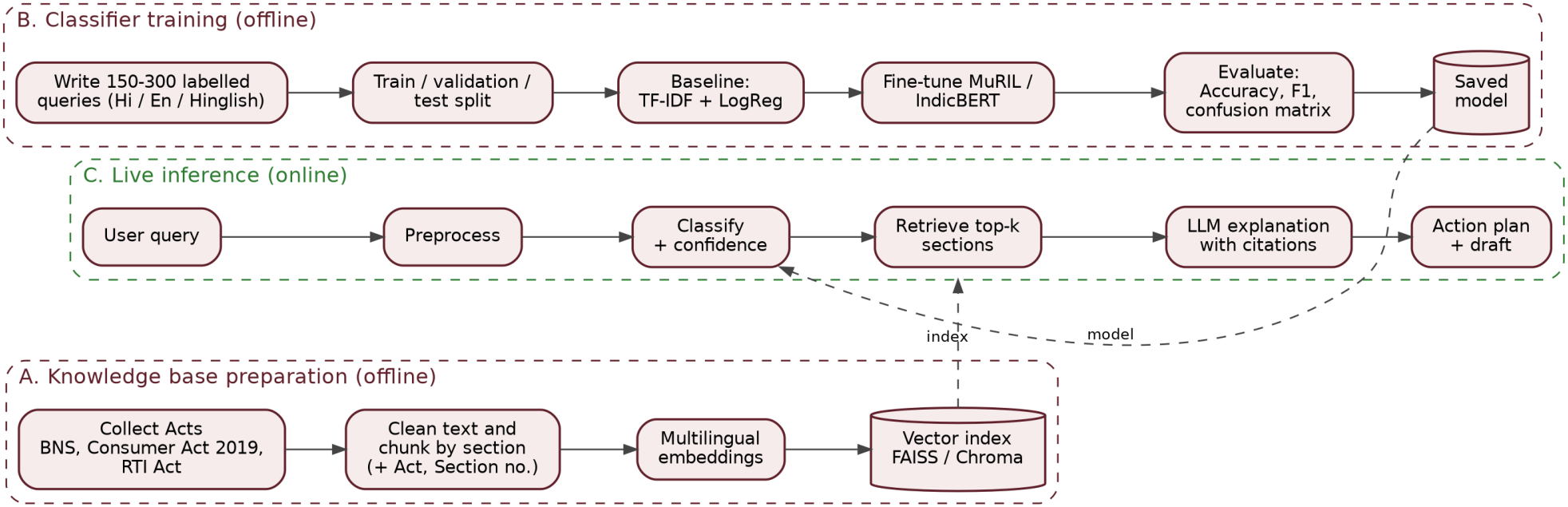


Fig. 8: Sequence Diagram

Message flow between the Citizen, Web UI, FastAPI backend, classifier, retriever, vector index, LLM generator and database for a query, with an alt frame for low confidence and an opt frame for draft generation.

